

Auteur: Mathijs Pittoors
 Studierichting: Informatica
 Oefeningenreeks 3H nr. 5.3

$$r = -\frac{3}{2} \cos \theta$$

Periode van 2π , we bekijken het interval $[0, 2\pi]$

Nulpunten r:

$$0 = -\frac{3}{2} \cos \theta$$

$$\Rightarrow 0 = \cos \theta$$

$$\Rightarrow \theta_1 = \frac{\pi}{2} \text{ en } \theta_2 = \frac{3\pi}{2}$$

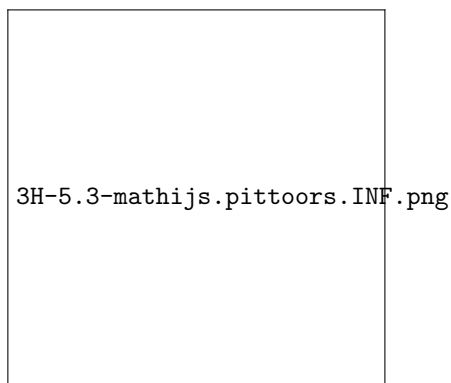
Nulpunten afgeleide r:

$$0 = \frac{3}{2} \sin \theta$$

$$\Rightarrow 0 = \sin \theta$$

$$\Rightarrow \theta_k = k\pi \quad \forall k \in \{0, 1, 2\}$$

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
r	−	0	+	0	−
r'	0	+	0	−	0
r	$\min\left(-\frac{3}{2}\right)$	$\nearrow$	$\max\left(\frac{3}{2}\right)$	$\searrow$	$\min\left(-\frac{3}{2}\right)$



References